MILITARY SPECIFICATION

DIVERS HELMET, UNDERWATER BREATHING APPARATUS MK-12

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and Is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements of the underwater breathing apparatus (UBA), Mark 12, Mod O Helmet.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the-issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

BB-A-1034 - Compressed Air, Breathing.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-comer.
PPP-B-636 - Boxes, Shipping, Fiberboard.

MILITARY

MIL-D-16791 - Detergents, General Purpose (Liquid, Nonionic).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use In improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end Of this document Or by letter.

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STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-454 - Standard General Requirements for Electronic Equipment.
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking and Waterproofing; With Appropriate Test Methods.
MIL-STD-1330 - Cleaning and Testing of Shipboard Oxygen, Nitrogen and Hydrogen Gas Piping Systems.

2.1.2 Government drawings. The following Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)
NAVSHIPS DL 592-4758500 - Helmet Assembly Data List
NAVSHIPS 592-4758593 - Non-Return Valve

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity-)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 Item composition. The Mark 12 Mod O Helmet is composed of all items specified on Drawing DL 592-4758500. The helmet is part of the MK-12 surface supported diving system (SSDS).

3.3 Characteristics.

3.3.1 Vibration. The helmet shall show no damage after being tested as specified in 4.6.2.

3.3.2 Performance. The helmet shall meet all performance requirements specified on Drawing DL 592-4758500 and 3.3.2.1 through 3.3.2.4.

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3.3.2.1 Nonreturn valve. The nonreturn valve specified on Drawing 592-4758593 shall be installed in the helmet and hold a minimum of 20-inch mercury vacuum for not less than 3 minutes with a loss of not more than 3-inch mercury vacuum.

3.3.2.2 Leaks. The blanked off helmet assembly, when fully immersed in water and with the supply and exhaust valves properply adjusted, shall hold an air pressure of 3 ± 0.4 pounds per square inch (lb/in^2) for 3 minutes without leaks.

3.3.2.3 Pressure and flow. The helmet assembly with not less than 40 lb/in^2 of air to the helmet and the supply valve adjusted to deliver 6 ± 0.5 actual cubic feet per minute (A ft^3/min) shall maintain a pressure of 2.0 ± 0.3 lb/in^2 with the exhaust valve closed.

3.3.2.4 Communications. The helmet communications component microphone and. speaker shall operate in a clear intelligible manner with variable volume when tested as specified in 4.6.3.3.

3.4 Construction. The helmet shall be constructed using the best commercial practices and procedures where such practices and procedures are not specifically specified on Drawings DL 592-4758500 and 592-4758593.

3.4.1 Production drawings. The helmet shall be fabricated and assembled in accordance with the drawings, parts lists, and other documents specified on Drawing DL 592-4758500.

3.4.2 Standards of manufacture. The helmet shall meet the requirements of all applicable drawings and specifications. No material shall be used that could cause a safety or health hazard.

3.5 Cleaning area. Components shall be cleaned in an area that is not conductive to hydrocarbon contamination, as specified in MIL-STD-1330.

3.5.1 Component cleaning methods. Components, except those noted, shall be cleaned prior to assembly into helmet subassemblies and assemblies. Metal components , viewports, and the helmet liner shall be washed in a warm water and mild, nonionic detergent solution as specified in MIL-D-16791 flushed with oil free, filtered warm water; and blown dry with air as specified in BB-A-1034, grade B or D which is free of oil vapor, gross moisture, dust, and other foreign matter. Softwear components shall be washed in isopropyl alcohol and blown dry with air as specified herein. Excepted components are:

Helmet - Earphone, microphone , air supply diffuser filter elements and air supply filter.

3.6 Workmanship. The helmet shall meet the workmanship requirements specified in MIL-STD-454, requirement 9.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved-by-the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the inspections and tests specified in 4.5 and 4.6.

4.3.1 Sampling for first article inspection. The contractor shall submit one Mark 12, Mod 0 Helmet for inspection. The first article shall be manufactured and packaged using methods, materials, and procedures identical to those proposed for production.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the inspections and test specified in 4.5 and 4.6.3.1 through 4.6.3.3.

4.5 Physical inspection. Parts, materials, and assemblies shall be inspected to determine conformance to all requirements of this specification and requirements of all drawings specified on Drawing DL 592-4758500.

4.6 Tests.

4.6.1 Test conditions. Unless otherwise specified, all tests shall be conducted at the ambient. conditions of the testing activity.

4.6.2 Vibration. The helmet, secured in its shipping container, (closed as required by the applicable container specification) shall be vibrated for 1 hour in each of the three major axis in accordance with MIL-STD-810, method 514.3, category 3. Upon completion of the test there shall be no damage to the helmet that interferes with its intended use.

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4.6.3 Operation performance. After successful completion of 4.5 and 4.6.2 the MK-12 helmet shall be subjected to the following operating performance testing:

4.6.3.1 Nonreturn valve. The nonreturn valve shall be tested as follows to ensure compliance with 3.3.7.]:

(a) Remove the protective cap from the helmet air adapter and the plug from the test set OUTLET adapter.
(b) Connect the vacuum gauge adapter to the helmet air adapter.
(c) Connect the vacuum pump and vacuum gauge to vacuum gauge adapter quick disconnects.
(d) Open helmet air supply valve at least one full turn.
(e) Operate the vacuum pump trigger until a 20-inch mercury (Hg) vacuum is indicated on the VACUUM gauge.
(f) Disconnect the vacuum pump from the test set.
(g) Record the VACUUM gauge reading after 1 minute. Failure of the nonreturn valve to meet the requirement specified in 3.3.2 shall be cause for rejection.
(h) Reconnect the vacuum pump and open the vent. Bleed off until the VACUUM gauge indicates zero.

4.6.3.2 Leak and pressure flow. The helmet shall be tested as follows to ensure compliance with 3.3.2.2 and 3.3.2.3. The inlet gas supply pressure shall be 40 lb/in^2 minimum.

(a) Install the helmet blanking plate and connect pressure O-5 lb/in^2 gauge to blanking plate quick disconnect.
(b) Remove the protective cap from the helmet air adapter.
(c) Connect the air whip to the helmet air adapter.
(d) Remove the protective plug from the umbilical hose and blow the hose down from the console.
(e) Connect umbilical hose to helmet air whip.
(f) Adjust console pressure to 100 lb/in^2. Open the helmet air supply valve approximately one-fourth turn to establish a flow of 6 ± 0.5 A ft^3/min.
(g) If helmet pressure is excessive, decrease the exhaust valve spring tension.
(h) With helmet exhaust valve fully open, adjust exhaust ring tension screw until a helmet pressure of 0.3 ± 0.05 lb/in^2 is achieved.
(i) Shut helmet exhaust valve and record helmet pressure. Failure to meet the requirement of 3.3.2.3 shall be cause for rejection.
(j) Leak test the helmet assembly.

10 Fully shut exhaust valve.
29 Fully submerge the helmet in water for 3 minutes with an air pressure of 3 ± 0.04 lb/in2.
3* Failure of the helmet to meet the requirement of 3.3.2.2 shall be cause for rejection.
4. Remove helmet from water.

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(k) open exhaust valve (#18).

(1) Shut test set supply valve and open BLEED valve until the SUPPLY gauge indicates O lb/in^2.

4.6.3.3 Communications. A communicator will be provided by the Government to permit an operational test of the helmet microphone and speakers after assembly to ensure compliance with 3.3.2.4. Connect the helmet whip connector to any of the three diver jacks on the front of the communicator. Turn the communicator on/off switch to the off position to prevent acoustic feedback from speaker to the helmet microphone. Turn the diver volume pot down. Power the communicator. Verify speech emanating from both helmet speakers, that speaker volume is adequate when the volume pot is raised to approximately mid-setting, and that speech is clear and Intelligible.

4.7 Inspection of packaging. Sample packages and packs, and the inspection of the preservation-packaging, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.5.)

5.1 General requirements.

5.1.1 lCleaning and drying. Cleaning and drying of the equipment components shall be as specified in 3.5 and 3.5.1.

5.1.2 Dusting powder. Talc or talcum used in the packaging process of items or furnished as a support item shall be free of asbestos.

5.2 Preservation.

5.2.l Helmet assembly. Each helmet assembly including the communications whip, less the helmet breech ring performed packing (O-ring), shall immediately after cleaning be packed in a transparent waterproof greaseproof plastic bag, not less than 6 roils thick. The communication whip shall be coiled and the end connector wrapped and cushioned to prevent damage to the helmet parts or plastic bag. The method of coiling the whip shall result in the most practical minimum package cube. The plastic bag including contents shall be purged with dry oil-Free nitrogen and the bag closed by heat sealing.

5.2.1.1 Breech ring preformed packing (O-ring). The preformed packing shall be packaged in its normal form or coiled in a "U"' or "S" shape so that it does not cross or lap over itself. Care shall be exercised in coiling the packing to avoid possible cold-crease effect. After coiling, the packing shall be sandwiched between two stiffeners of fiberboard box material. The stiffeners shall be fastened by taping with pressure-sensitive tape on the four edges. The fiberboard shall not be compressed in such a manner which would deform the packing; however, the packing shall not be permitted to uncoil nor come in contact with pressure sensitive tape. The sandwiched packing shall then be placed in an opaque waterproof greaseproof bag. The bag closure shall be by heat sealing.

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5.3 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.3.1 Levels A and B.

5.3.1.1 Helmet assembly and preformed packing. The helmet assembly and breech ring preformed-packing preserved-packaged as specified shall be packed in a weather-resistant fiberboard container conforming to PPP-B-636 special requirements, grade V with variety and type optional. Blocking and cushioning shall be applied to effect a type 1 or 2 load as defined in the appendix to PPP-B-636. Die cuts, partitions, or other fiberboard used shall be of the same material as the container. Container closure shall conform to method V and reinforcing shall be with nonmetallic banding or tape as specified in the appendix to PPP-B-636.

5.3.2 Consolidation. When more than one helmet is to be furnished, the packed items (see 5.3.1.1) shall be overpacked in a wood or wood-cleated plywood box conforming to PPP-B-621 overseas class 2 or PPP-B-601 overseas type respectively. Containers shall be closed, strapped, *or* banded in accordance with the applicable container specification or appendix thereto. Quantity of helmets per container shall be 2, 3, or 6.

5.3.3 Cushioning, anchoring, blocking and bracing. Cushioning, anchoring, blocking, and bracing of the container contents shall be in accordance with MIL-STD-1186, the applicable container specification and the appendix thereto.

5.3.4 Cushioning filler, dunnage, and wrapping materials.

5.3.4.1 Level A preservation-packaging and levels A and B packing. Use of all types of loose-fill materials for packaging and packing applications such as cushioning , filler, or dunnage is prohibited for materials destined for shipboard installation and or storage.

5.3.4.2 Level C preservation-packaging and packing. When loose fill type materials are used for packaging and packing applications such as cushioning, filler and dunnage, all containers (unit, intermediate, and shipping) shall be marked or labeled with the following information.

"'CAUTION

Contents cushioned with loose-fill material shall not be taken on board ship. Remove and discard loose-fill material. If required, recushion with cellulosic material, bound fiber, fiberboard, or transparent flexible cellular material,"

5.3.4.3 Cushioning, filler, dunnage and wrapping materials selected; whenever available, shall exhibit improved performance for resistance to fire.

5.3.5 Level C. The helmet assembly shall be packed as specified in 5.3.1.1 and require no further packing.

5.4 Marking. In addition to any special markings required (see 6.2), interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

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5.4.1 Arrows. All shipping containers shall be marked with the word "UP" and an arrow pointing to the container top in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The Mark 12, Mod O Helmet assembly is intended for use as a component of the diving system (MK-12 SSDS) which will be used to support all surface-supported diving requirements of the U.S. Navy.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) If first article inspection is required (see 3.1).
(c) Level of preservation, packaging, and packing required (ace 5.2 and 5.3).
(d) Degree of marking required (see 5.4).

6.3 First article inspection. When a first article inspection is required, the item should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Subject term (key word) listing.

Breathing apparatus
Divers helmet
Earphone
Microphone
Speaker
Valves

6.5 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

Preparing activity:
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-24662(SH)	DIVERS HELMET, UNDERWATER BREATHING APPARATUS MK-12

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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